

Lista 2

$$\textcircled{1} f(x) = x^8 - 2x^7 + 5x^4 - 10x^2 + 20$$

$$\varepsilon = 10^{-2}$$

$$\therefore f'(1) \text{ e } f''(1)$$

$$f'(x) = \frac{f(x+h) - f(x-h)}{2h} \quad \Delta$$

$$\begin{array}{l} i=0 \\ h=1 \end{array}$$

$$f'(1) = \frac{f(1+1) - f(1-1)}{2} = \underline{20}$$

$$\begin{array}{l} i=1 \\ h=0,5 \end{array}$$

$$f'(1) = \frac{f(1+0,5) - f(1-0,5)}{1} = 14,2695 - 17,8007 = \underline{-3,5312}$$

$$\begin{array}{l} i=2 \\ h=0,25 \end{array}$$

$$f'(1) = \frac{f(1+0,25) - f(1-0,25)}{\frac{1}{2}} = 2 \cdot (13,0057 - 15,7901) = \underline{-5,5688}$$

$$\begin{array}{l} i=3 \\ h=0,1250 \end{array}$$

$$f'(1) = \frac{f(1+0,125) - f(1-0,125)}{0,25} = \underline{-5,9028}$$

$$\begin{array}{l} i=4 \\ h=0,0625 \end{array}$$

$$f'(1) = \frac{f(1+0,0625) - f(1-0,0625)}{\frac{1}{8}} = \underline{-5,9763}$$

$$\begin{array}{l} i=5 \\ h=\frac{1}{16} \end{array}$$

$$f'(1) = \frac{f(1+\frac{1}{16}) - f(1-\frac{1}{16})}{\frac{1}{16}} = \underline{-5,9941}$$

$$\text{Err} = \frac{|-5,9941 + 5,9763|}{|-5,9941|} = \underline{0,00296}$$

$$\underline{f'(1) = -5,9941}$$

$$f''(1) = ?$$

$$\lambda = 0$$

$$h = 1$$

$$f''(1) = \frac{f(1+1) - 2 \cdot f(1) + f(1-1)}{1}$$

$$f''(1) = \frac{60 - 28 + 20}{1} = \underline{52}$$

$$\lambda = 1$$

$$h = 0.5$$

$$f''(1) = \frac{f(1.5) - 2 \cdot f(1) + f(0.5)}{\left(\frac{1}{2}\right)^2} = 16.2812$$

$$\lambda = 2$$

$$h = \frac{1}{4}$$

$$f''(1) = \frac{f(1.25) - 2 \cdot f(1) + f(0.75)}{\left(\frac{1}{4}\right)^2} = 12.7348$$

$$\lambda = 3$$

$$h = \frac{1}{8}$$

$$f''(1) = \frac{f(1.125) - 2 \cdot f(1) + f(0.875)}{\left(\frac{1}{8}\right)^2} = 12.1630$$

$$\lambda = 4$$

$$h = \frac{1}{16}$$

$$f''(1) = \frac{f(1.0625) - 2 \cdot f(1) + f(0.9375)}{\left(\frac{1}{16}\right)^2} = 12.03949$$

$$\lambda = 5$$

$$h = \frac{1}{32}$$

$$f''(1) = \frac{f(1.0312) - 2 \cdot f(1) + f(0.9687)}{\left(\frac{1}{32}\right)^2} = 12.00979$$

$$Err = \frac{|12.0097 - 12.0394|}{12.0097} = 0.00247$$

⑥ sec²(x)

$i=0$
 $h=1$

$$f''(\pi/3) = \frac{f(\pi/3+1) - f(\pi/3-1)}{2} = 0.3937$$

$i=1$
 $h=0.5$

$$f'(\pi/3) = \frac{f(\pi/3 + \frac{1}{2}) - f(\pi/3 - \frac{1}{2})}{1} = 0.7287$$

$i=2$
 $h=0.25$

$$f'(\pi/3) = \frac{f(\pi/3 + \frac{1}{4}) - f(\pi/3 - \frac{1}{4})}{\frac{1}{2}} = 0.8303$$

> Err > ε

$i=3$
 $h=0.125$

$$f'(\pi/3) = \frac{f(\pi/3 + \frac{1}{8}) - f(\pi/3 - \frac{1}{8})}{\frac{1}{4}} = 0.8570$$

> Err > ε

$i=4$
 $h=0.0625$

$$f'(\pi/3) = \frac{f(\pi/3 + \frac{1}{16}) - f(\pi/3 - \frac{1}{16})}{\frac{1}{8}} = 0.8637$$

> Err < ε

$f'(\pi/3) \approx 0.8637$

$f''(\pi/3) \approx -1$

$x = \pi/3$

$i=0$
 $h=1$

$$f''(\pi/3) = \frac{f(\pi/3+1) - 2f(\pi/3) + f(\pi/3-1)}{(1)^2} = -0.7080$$

→ Err > ε

$i=1$
 $h=0.5$

$$f''(x) = \frac{f(x+0.5) - 2f(x) + f(x-0.5)}{(\frac{1}{2})^2} = -0.9194$$

$i=2$
 $h=0.25$

$$f''(x) = \frac{f(x+0.25) - 2f(x) + f(x-0.25)}{(\frac{1}{4})^2} = -0.9793$$

→ Err > ε

$$\lambda = 3$$

$$h = 0.125$$

$$f''(x) = \frac{f(x+0.125) - 2 \cdot f(x) + f(x-0.125)}{\left(\frac{1}{8}\right)^2}$$

$$= \underline{\underline{-0.9948}}$$

$$\lambda = 4$$

$$h = 0.0625$$

$$f''(x) = \frac{f(x+\frac{1}{16}) - 2 \cdot f(x) + f(x-\frac{1}{16})}{\left(\frac{1}{16}\right)^2}$$

$$= \underline{\underline{-0.9987}}$$

$$Err = \frac{|-0.9987 + 0.9948|}{1} = \underline{\underline{0.0039}} < \epsilon$$

$$f''(\pi/3) \approx \underline{\underline{-0.9987}}$$

③ $f(x) = (\sin(x) \cdot \cos(3x))^2$, $f'(-2) = ?$ e $f''(1.5) = ?$

$$\lambda = 0$$

$$h = 1$$

$$f'(x) = \frac{f(-2+1) - f(-2-1)}{2} = \underline{\underline{0.3387}}$$

$\Rightarrow Err > \epsilon$

$$\lambda = 1$$

$$h = 0.5$$

$$f'(x) = \frac{f(-2+0.5) - f(-2-0.5)}{1} = \underline{\underline{0.0011}}$$

$$\lambda = 2$$

$$h = 0.25$$

$$f'(x) = \frac{f(-2+0.25) - f(-2-0.25)}{\left(\frac{1}{2}\right)} = \underline{\underline{-0.5890}}$$

$$\lambda = 3$$

$$h = 0.125$$

$$f'(x) = \frac{f(-2+0.125) - f(-2-0.125)}{\frac{1}{4}} = \underline{\underline{-0.6222}}$$

$$i=4$$

$$h=0.0625$$

$$f'(x) = \frac{f(-2 + \frac{1}{6}) - f(-2 - \frac{1}{6})}{\frac{1}{8}}$$

$$= \underline{\underline{-0.6222}}$$

$$i=5$$

$$h=0.03125$$

$$f'(x) = \frac{f(-2 + \frac{1}{32}) - f(-2 - \frac{1}{32})}{\frac{1}{16}}$$

$$= \underline{\underline{-0.63047}}$$

$$f'(x) \cong -0.6304 \quad \therefore \text{Err} = |-0.6304 + 0.6222|$$

$$f''(x) = ?$$

$$\text{Err} < \varepsilon$$

$$\begin{array}{r} 0.6304 \\ - 0.6222 \\ \hline 0.0082 \end{array}$$

$$i=0$$

$$h=1$$

$$f''(x) = \frac{f(1.5+1) - 2 \cdot f(1.5) + f(1.5-1)}{(1)^2}$$

$$= \underline{\underline{-0.0424}}$$

$$i=1$$

$$h=0.5$$

$$f''(1.5) = \frac{f(1.5+0.5) - 2 \cdot f(1.5) + f(1.5-0.5)}{(\frac{1}{2})^2}$$

$$= \underline{\underline{5.4712}}$$

$$i=2$$

$$h=0.25$$

$$f''(1.5) = \frac{f(1.5+0.25) - 2 \cdot f(1.5) + f(1.5-0.25)}{(\frac{1}{4})^2}$$

$$= \underline{\underline{12.3495}}$$

$$\rightarrow \text{Err} > \varepsilon$$

$$i=3$$

$$h=0.125$$

$$f''(1.5) = \frac{f(1.5+0.125) - 2 \cdot f(1.5) + f(1.5-0.125)}{(\frac{1}{8})^2}$$

$$= \underline{\underline{14.3495}}$$

$$i = 4$$

$$h = 0.0625$$

$$f''(x) = \frac{f(x + \frac{1}{16}) - 2 \cdot f(x) + f(x - \frac{1}{16})}{(\frac{1}{16})^2}$$

$$f''(x) = 15.6375$$

$$i = 5$$

$$h = 0.0312$$

$$f''(x) = \frac{f(x + \frac{1}{32}) - 2 \cdot f(x) + f(x - \frac{1}{32})}{(\frac{1}{32})^2}$$

$$f''(x) = 15.8200$$

$$Err = \frac{|15.8200 - 15.6375|}{15.8200} = 0.1154$$

$$\therefore Err < \epsilon$$

$$i = 6$$

$$h = 0.0156$$

$$f''(x) = \frac{f(x + \frac{1}{64}) - 2 \cdot f(x) + f(x - \frac{1}{64})}{(\frac{1}{64})^2}$$

$$f''(1.5) = 15.8660$$

$$\therefore Err < \epsilon$$

(d) ~~$x_1^2 + 2x_2^3$~~ , $\frac{\partial f}{\partial x_1}$, $x = \begin{pmatrix} 2 \\ -2 \end{pmatrix}$

$$i = 1$$

$$h = 1$$

$$\frac{\partial f}{\partial x_1} = \frac{f(2+1, -2) - f(2-1, -2)}{2} = 4$$

$$i = 2$$

$$h = \frac{1}{2}$$

$$\frac{\partial f}{\partial x_1} = \frac{f(2+\frac{1}{2}, -2) - f(2-\frac{1}{2}, -2)}{1} = 4$$

$$Err = \frac{|4 - 4|}{4} = 0 < \epsilon \therefore \frac{\partial f}{\partial x_1} = 4$$

© $f(x) = \frac{x_1 + x_2^2}{x_1 x_2}$, Calcule $\frac{\partial f}{\partial x_2}$ no ponto $x = \begin{pmatrix} -2 \\ 2 \end{pmatrix}$

$$\begin{matrix} i=1 \\ h=1 \end{matrix} \quad \frac{\partial f}{\partial x_2} = \frac{f(-2, 2+1) - f(-2, 2-1)}{2} = -0,8333$$

$$\begin{matrix} i=2 \\ h=1/2 \end{matrix} \quad \frac{\partial f}{\partial x_2} = \frac{f(-2, 2+1/2) - f(-2, 2-1/2)}{1} = -0,7667$$

$$Err = -1,5997 > \epsilon$$

$$\begin{matrix} i=3 \\ h=1/4 \end{matrix} \quad \frac{\partial f}{\partial x_2} = \frac{f(-2, 2+1/4) - f(-2, 2-1/4)}{1/2} = -0,7540$$

$$\begin{matrix} i=4 \\ h=1/8 \end{matrix} \quad \frac{\partial f}{\partial x_2} = \frac{f(-2, 2+1/8) - f(-2, 2-1/8)}{1/8} = -0,7540$$

$$Err = 3 \cdot 10^{-3} \therefore Err < \epsilon$$

$$\frac{\partial f}{\partial x_2} = -0,7540$$

$$f) \quad f(x) = 2x_1 - 2x_1x_2 + x_2^2 \quad \nabla f(x) \text{ no ponto } x = \begin{pmatrix} 7 \\ 3 \end{pmatrix}$$

$$i) \text{ calc } \frac{\partial f}{\partial x_1}$$

$$\begin{matrix} i=1 \\ h=1 \end{matrix} \quad \frac{f(7+1, 3) - f(7-1, 3)}{2} = -4$$

$$\begin{matrix} i=2 \\ h=1/2 \end{matrix} \quad \frac{f(7+1/2, 3) - f(7-1/2, 3)}{1} = -4$$

$$\text{Err} = 0 < \varepsilon \therefore \frac{\partial f}{\partial x_1} = -4$$

$$ii) \rightarrow \frac{\partial f}{\partial x_2}$$

$$\begin{matrix} i=1 \\ h=1 \end{matrix} \quad \frac{\partial f}{\partial x_2} = \frac{f(7, 3+1) - f(7, 3-1)}{2} = -8$$

$$\begin{matrix} i=2 \\ h=1/2 \end{matrix} \quad \frac{\partial f}{\partial x_2} = \frac{f(7, 3+1/2) - f(7, 3-1/2)}{1} = -8$$

$$\text{Err} = 0 \therefore \frac{\partial f}{\partial x_2} = -8 \quad \nabla f(7, 3) = \begin{pmatrix} -4 \\ -8 \end{pmatrix}$$

g) $f(x) = 3x_1 + 2x_1x_2 + x_2^3$ Param $x = 1/4$

i)

$$i=1, h=1 \quad \frac{\partial f}{\partial x_2} = \frac{f(1+1, 4) - f(1-1, 4)}{2} = 11$$

$$i=2, h=1/2 \quad \frac{\partial f}{\partial x_2} = \frac{f(1+1/2, 4) - f(1-1/2, 4)}{1} = 11$$

$$\text{Err} = 0 < \epsilon \quad \therefore \frac{\partial f}{\partial x_2} = 11$$

ii)

$$i=1, h=1 \quad \frac{\partial f}{\partial x_2} = \frac{f(1, 4+1) - f(1, 4-1)}{2} = \underline{\underline{51}}$$

$$i=2, h=1/2 \quad \frac{\partial f}{\partial x_2} = \frac{f(1, 4+1/2) - f(1, 4-1/2)}{1} = 50,25$$

$$\text{Err} = \frac{0,00497}{-0,014} > \epsilon \quad \therefore$$

$$i=3, h=1/4 \quad \frac{\partial f}{\partial x_2} = \frac{f(1, 4+1/4) - f(1, 4-1/4)}{1/2} = 50,0625$$

$$\text{Err} < \epsilon \quad \therefore \nabla f = \begin{pmatrix} 11 \\ 50,0625 \end{pmatrix}$$

h)

$\sigma(x)$

$$x = \begin{pmatrix} 1 \\ 5 \end{pmatrix}$$

$$\bullet f_1(x) = 2x_1 - 2x_1x_2 + x_2^2$$

$$\bullet f_2(x) = x_1^2 + 2x_2^3$$

$$\bullet f_3(x) = 5x_1^3 + 2x_1x_2$$

1)

i)

$\frac{\partial f}{\partial x_1}(x)$

$i=1$
 $h=1$

$$\frac{\partial f}{\partial x_1} = \frac{f(1+1, 5) - f(1-1, 5)}{2} = -8$$

$i=2$
 $h=1/2$

$$\frac{\partial f}{\partial x_1} = \frac{f(1+1/2, 5) - f(1-1/2, 5)}{1} = -8$$

Err = 0 < ε $\therefore \frac{\partial f}{\partial x_1} = -8$

ii)

$i=2$
 $h=1$

$$\frac{\partial f}{\partial x_2} = \frac{f(1, 5+1) - f(1, 5-1)}{2} = 8$$

$i=2$
 $h=1/2$

$$\frac{\partial f}{\partial x_2} = \frac{f(1, 5+1/2) - f(1, 5-1/2)}{1} = 8$$

Err = 0 < ε, então $\nabla f = \begin{pmatrix} -8 \\ 8 \end{pmatrix}$

2) $f_2(x)$

$i=1$
 $h=1$

$$\frac{f(1+1, 5) - f(1-1, 5)}{2} = 2$$

$i=2$
 $h=1$

$$\frac{f(1+1/2, 5) - f(1-1/2, 5)}{1} = 2$$

$\partial f = ?$

②

$$\begin{matrix} i=1 \\ n=1 \end{matrix} \quad \frac{f(1, 5+1) - f(1, 5-1)}{2} = 152$$

$$\begin{matrix} i=2 \\ n=1/2 \end{matrix} \quad \frac{f(1, 5+1/2) - f(1, 5-1/2)}{1} = 150,5$$

$$\begin{matrix} i=3 \\ n=1/4 \end{matrix} \quad \frac{f(1, 5+1/4) - f(1, 5-1/4)}{1/2} = \cancel{150,5} \quad \underline{150,125}$$

$$\text{Err} = 0,0025 \epsilon$$

$$\nabla f = \begin{pmatrix} 2 \\ 150,125 \end{pmatrix}$$

③

i) $f_3(x)$

$$\begin{matrix} i=1 \\ n=1 \end{matrix} \quad \frac{f(1+1, 5) - f(1-1, 5)}{2} = 30$$

$$\begin{matrix} i=2 \\ n=1/2 \end{matrix} \quad \frac{f(1+1/2, 5) - f(1-1/2, 5)}{1} = 26,25$$

$$\begin{matrix} i=3 \\ n=1/4 \end{matrix} \quad \frac{f(1+1/4, 5) - f(1-1/4, 5)}{1/2} = 25,3125$$

$$\text{Err} \geq \epsilon$$

$$\begin{matrix} i=4 \\ n=1/8 \end{matrix} \quad \frac{f(1+1/8, 5) - f(1-1/8, 5)}{1/4} = 25,078125$$

$$\text{Err} < \epsilon \therefore \frac{\partial f_3}{\partial x_1} = 25,078125$$

$$(ii) f_3(x_2)$$

$$\begin{matrix} i=1 \\ h=1 \end{matrix} \quad \frac{f(1, 5+1) - f(1, 5-1)}{2} = 2$$

$$\begin{matrix} i=2 \\ h=1/2 \end{matrix} \quad \frac{f(1, 5+1/2) - f(1, 5-1/2)}{1} = 2$$

$$\text{Err} \approx 0 < \varepsilon \quad \therefore \frac{\partial f_3}{\partial x_2} = 2 \quad \nabla f_3 = \begin{pmatrix} 25,0784 \\ 2 \end{pmatrix}$$

$$\nabla f(x) =$$

$$J(x) = \begin{pmatrix} -8 & 8 \\ 2 & 150,125 \\ 25,0784 & 2 \end{pmatrix}$$